

CHP#17: SELECTION CONTROL STRUCTURES

- **Topics Which We Will Cover in this Chapter:**

- Control Structure A/C to Selection or Conditional Flow
- If, If-else, Nested If, Nested If-else Statements
- Switch-case Statements

Topic#01:

Control Structure A/C to Selection or Conditional Flow

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- **Control Structure:**

Control structures in C are constructs that determine the flow of control in a program. They allow you to specify conditions for executing different blocks of code, enabling complex decision-making and repetition.

- **Basic Control Structure:**

1. Sequence Control Structure
2. Selection Control Structure
3. Iteration Control Structure

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- **Selection Control Structure:**

Selection statements allow the program to choose different paths of execution based on conditions. They enable decision-making in the code.

- **Types of Selection Control Structure:**

1. If statements
2. If else statements
3. Nested if statements
4. Nested if else statements
5. Switch statements

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Any Doubt?

Topic#02:

If, If-else, Nested If, Nested If-else
Statements

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- **If statements:**

The if statement executes a block of code only if a specified condition is true.

- **Syntax:**

```
if (condition) {  
    // Code to be executed if the condition is true  
}
```

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- **Syntax:**

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}
```

- **Example:**

```
#include <stdio.h>  
  
int main() {  
    int num = 10;  
  
    if (num > 0) {  
        printf("The number is positive.\n");  
    }  
  
    return 0;  
}
```

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- **Syntax:**

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}
```

- **Example:**

```
#include <stdio.h>  
  
int main() {  
    int num = 10;  
  
    if (num > 0) {  
        printf("The number is positive.\n");  
    }  
  
    return 0;  
}
```

Output:
>>The number is positive.

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- **If-else statements:**

The if-else statement provides an alternative path of execution if the condition is false.

Syntax:

```
if (condition) {  
    // Code to be executed if the condition is true  
} else {  
    // Code to be executed if the condition is false  
}
```

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- **If-else statements:**

The if-else statement provides an alternative path of execution if the condition is false.

Syntax:

Example:

```
#include <stdio.h>

int main() {
    int num = -10;

    if (num > 0) {
        printf("The number is positive.\n");
    } else {
        printf("The number is negative or zero.\n");
    }

    return 0;
}
```

```
if (condition) {
    // Code to be executed if the condition is true
} else {
    // Code to be executed if the condition is false
}
```

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- **If-else statements:**

The if-else statement provides an alternative path of execution if the condition is false.

Syntax:

Example:

```
#include <stdio.h>

int main() {
    int num = -10;

    if (num > 0) {
        printf("The number is positive.\n");
    } else {
        printf("The number is negative or zero.\n");
    }

    return 0;
}
```

```
if (condition) {
    // Code to be executed if the condition is true
} else {
    // Code to be executed if the condition is false
}
```

Output:
>>The number is negative or zero.

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- **Homework Task#:**
- Try Nested If statements & Nested If-else statements.

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Any Doubt?

Topic#03:

Switch-case Statements

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- **Switch Statements:**

The switch statement selects one of many code blocks to be executed.

Syntax:

```
switch (expression) {  
    case constant1:  
        // Code to be executed if expression equals constant1  
        break;  
    case constant2:  
        // Code to be executed if expression equals constant2  
        break;  
    // You can have any number of case statements  
    default:  
        // Code to be executed if expression doesn't match any case  
}
```

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Any Doubt?